FALCUTÉ DES SCIENCES ET TECHNOLOGIES (FST)

TROISIÈME ANNÉE

Rapport du travail de Laboratoire N^{0} 1

Cours : Reseaux

Étudiant : BELDOUETTE Darvens Mardy Sharkyl

Professeur : Ismaël SAINT AMOUR

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INTRODUCTION

Ce travail dirigé porte sur l'adressage IPv4 et IPv6 dans un réseau informatique à l'aide de Cisco Packet Tracer. Il consiste à reproduire des topologies, attribuer des adresses IP aux équipements, puis vérifier la communication entre les machines à l'aide de la commande ping.

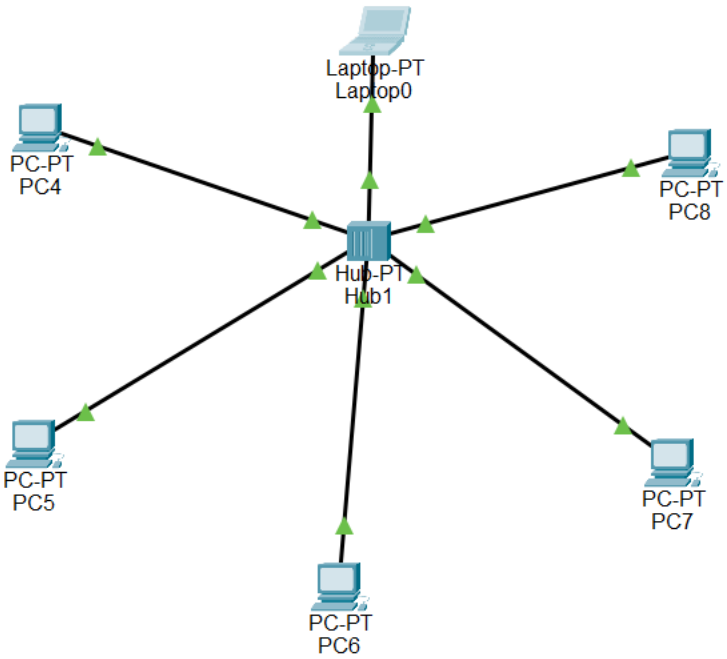
Objectifs du TD

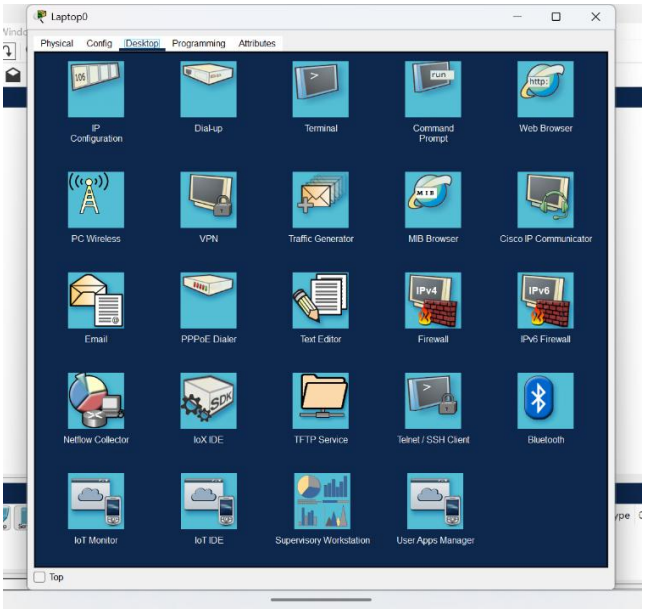
• Attribuer des adresses IP valides aux équipements du réseau.
• Comprendre les principes de l'adressage IPv4 et IPv6.
• Configurer les adresses IP sur les hôtes dans Cisco Packet Tracer.
• Vérifier la connectivité entre les machines avec les commandes ping IPv4 et IPv6.

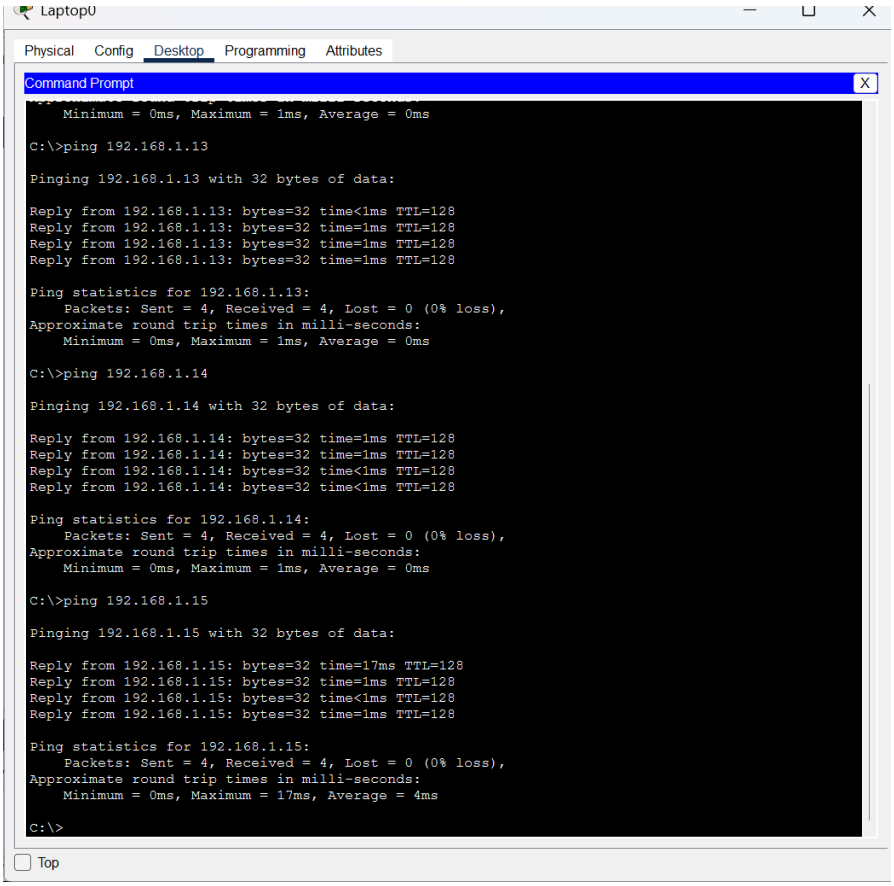
Méthodologie générale

Pour chaque exercice, la topologie demandée est reproduite dans Cisco Packet Tracer. Les équipements sont reliés conformément au schéma de l'énoncé. Les adresses IP indiquées dans le tableau de l'exercice sont ensuite configurées sur les interfaces des machines. Enfin, des tests de connectivité sont effectués avec la commande ping. Les captures d'écran des topologies, des configurations et des résultats doivent être ajoutées dans les espaces réservés de ce rapport.

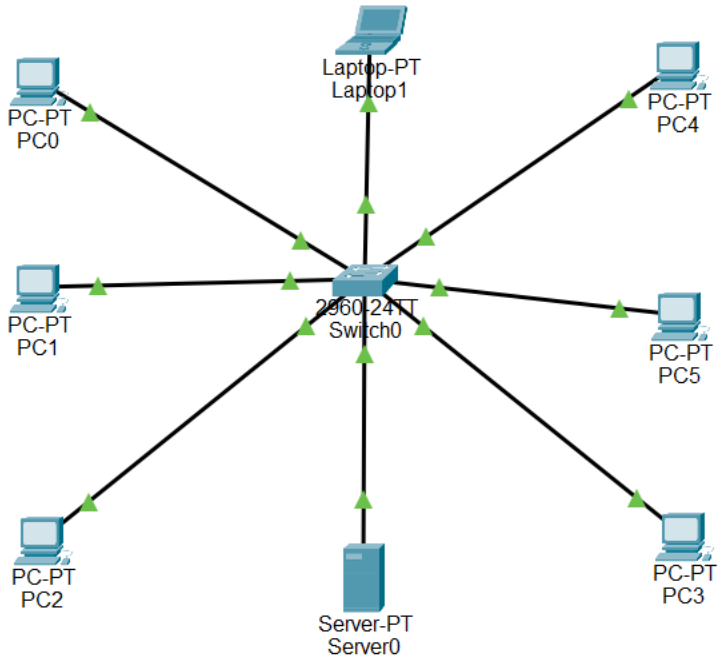
EXERCICE 1 – Configuration IPv4 – Topologie avec Hub



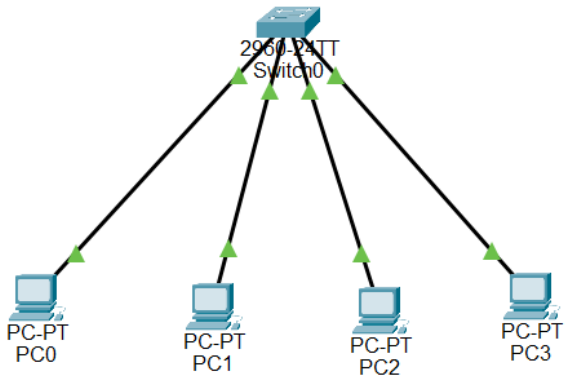

Laptop0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Dial-up
Terminal
Command Prompt
Web Browser
PC Wireless
VPN
Traffic Generator
MIB Browser
Cisco IP Communicator
Email
PPPoE Dialer
Text Editor
Firewall
IPv6 Firewall
Netflow Collector
IoX IDE
TFTP Service
Telnet / SSH Client
Bluetooth
IoT Monitor
IoT IDE
Supervisory Workstation
User Apps Manager
Top

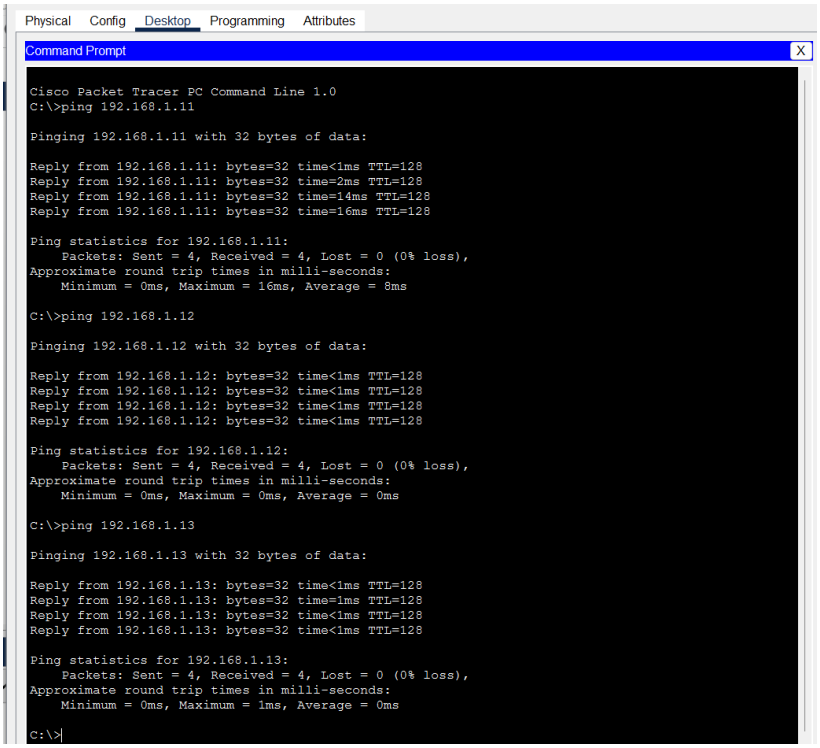

Laptop0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.1.13
Pinging 192.168.1.13 with 32 bytes of data:
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time=1ms TTL=128
Reply from 192.168.1.13: bytes=32 time=1ms TTL=128
Reply from 192.168.1.13: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.13:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.1.14
Pinging 192.168.1.14 with 32 bytes of data:
Reply from 192.168.1.14: bytes=32 time=1ms TTL=128
Reply from 192.168.1.14: bytes=32 time=1ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.14:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.1.15
Pinging 192.168.1.15 with 32 bytes of data:
Reply from 192.168.1.15: bytes=32 time=17ms TTL=128
Reply from 192.168.1.15: bytes=32 time=1ms TTL=128
Reply from 192.168.1.15: bytes=32 time<1ms TTL=128
Reply from 192.168.1.15: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.15:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 17ms, Average = 4ms
C:\>
Top

EXERCICE 2 – Configuration IPv4 – Topologie avec Switch

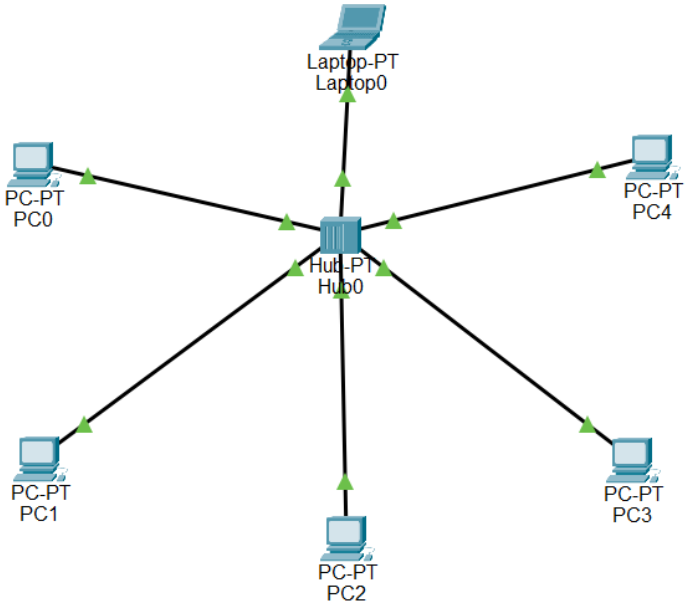


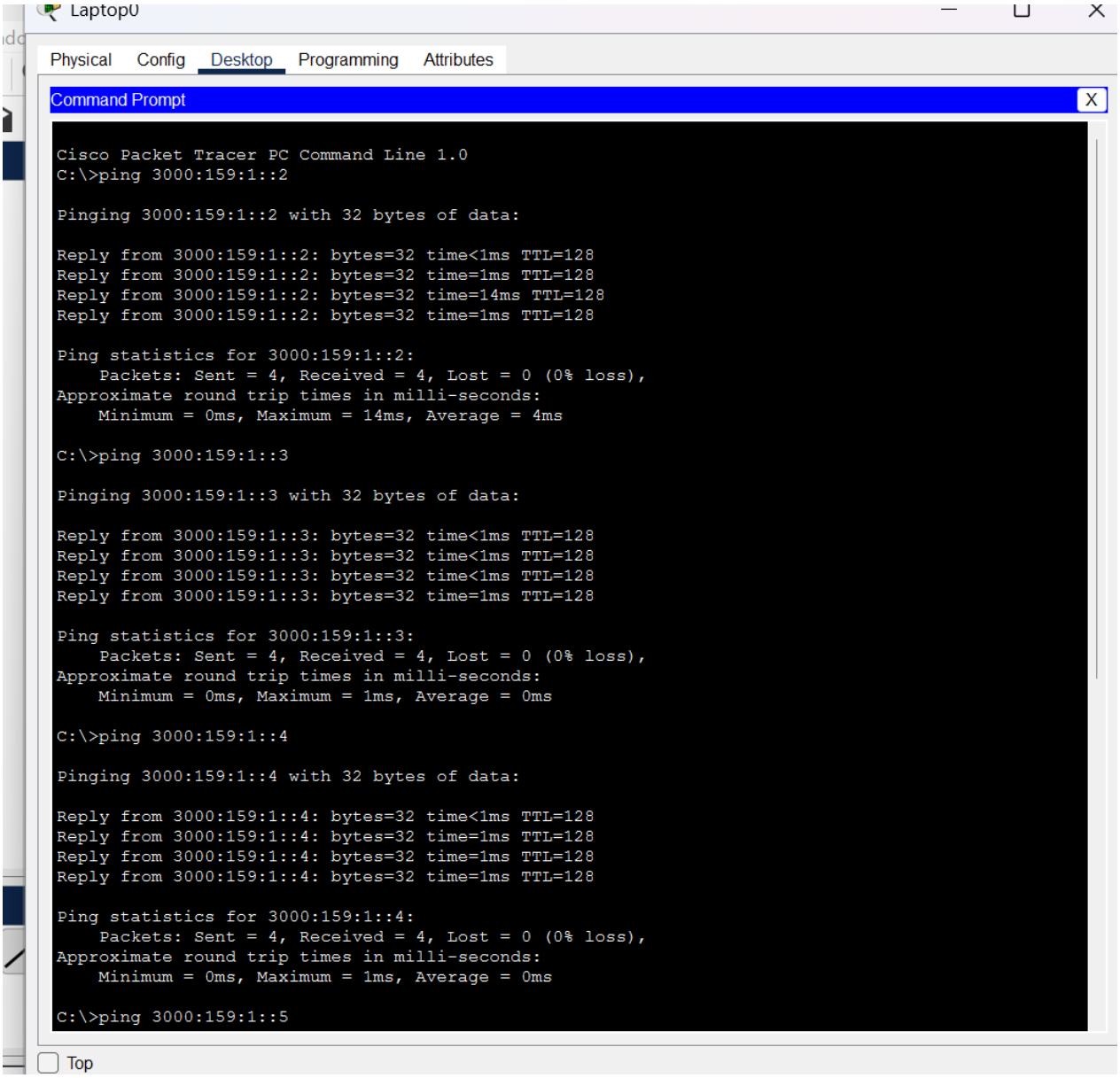
EXERCICE 3 – Choix d'une topologie – Configuration IP



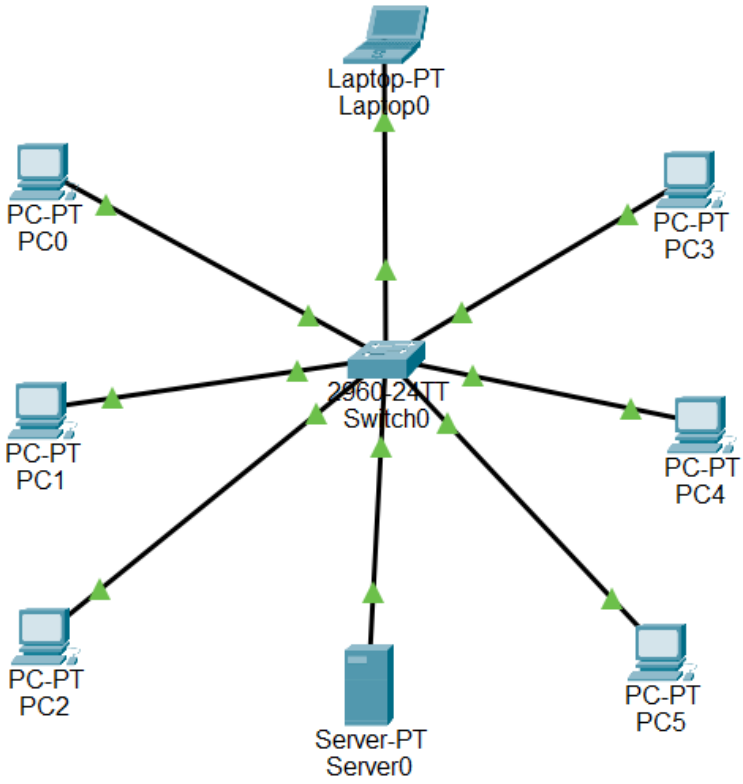


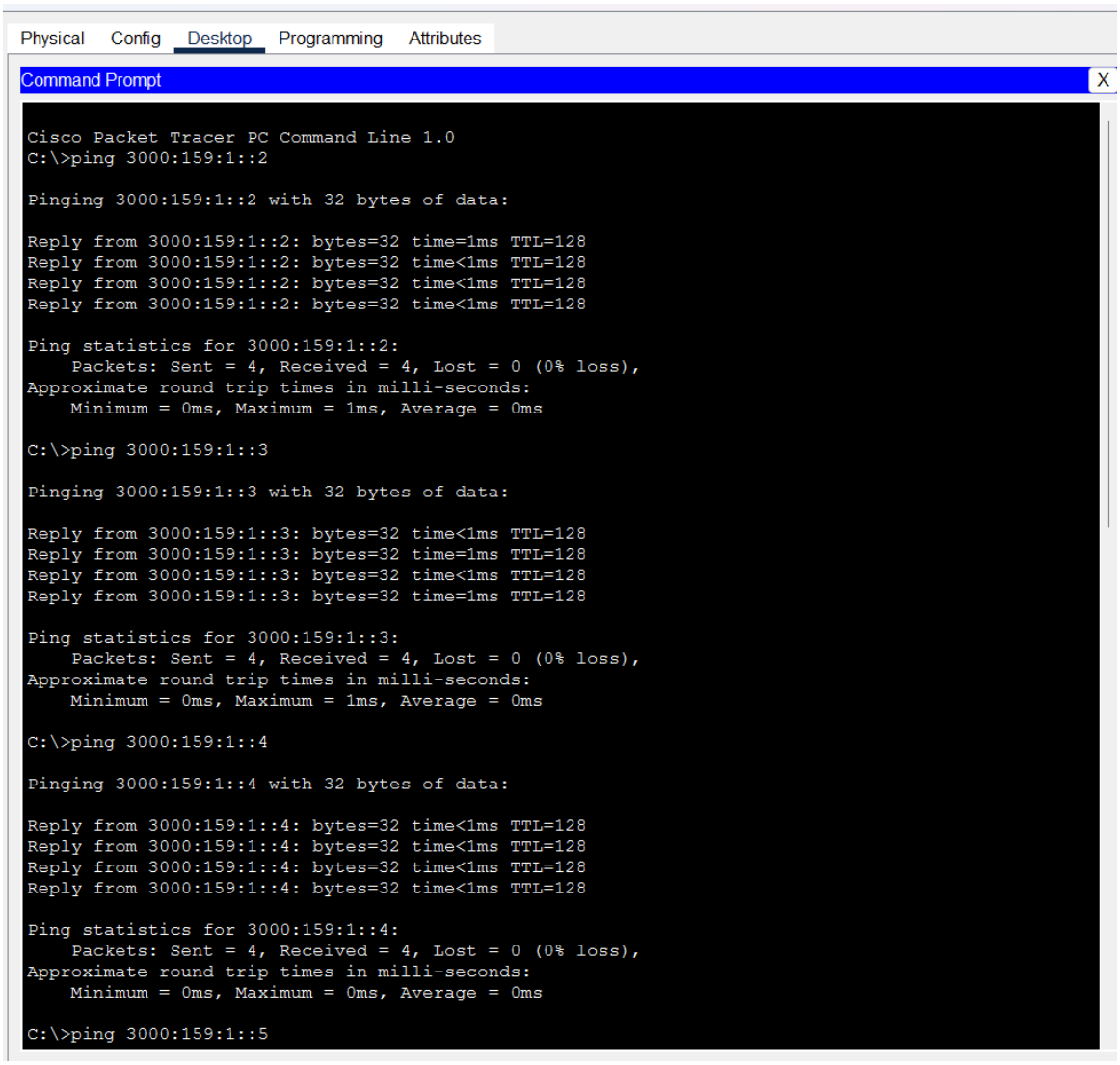
EXERCICE 4 – Configuration IPv6 – Topologie avec Hub




Laptop0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 3000:159:1::2
Pinging 3000:159:1::2 with 32 bytes of data:
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time=1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time=14ms TTL=128
Reply from 3000:159:1::2: bytes=32 time=1ms TTL=128
Ping statistics for 3000:159:1::2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 14ms, Average = 4ms
C:\>ping 3000:159:1::3
Pinging 3000:159:1::3 with 32 bytes of data:
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time=1ms TTL=128
Ping statistics for 3000:159:1::3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 3000:159:1::4
Pinging 3000:159:1::4 with 32 bytes of data:
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time=1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time=1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time=1ms TTL=128
Ping statistics for 3000:159:1::4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 3000:159:1::5
Top

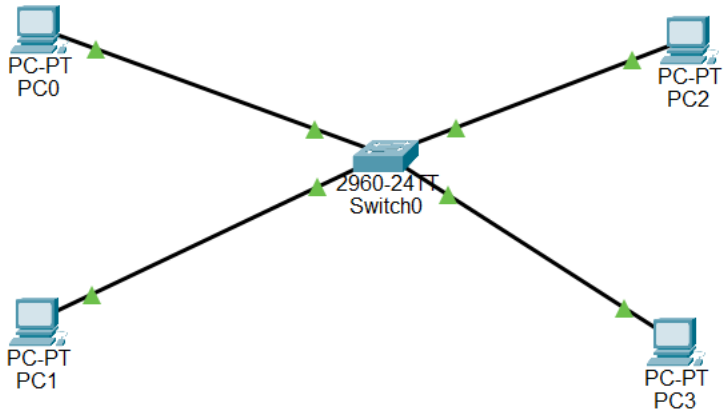
EXERCICE 5 – Configuration IPv6 – Topologie avec Switch

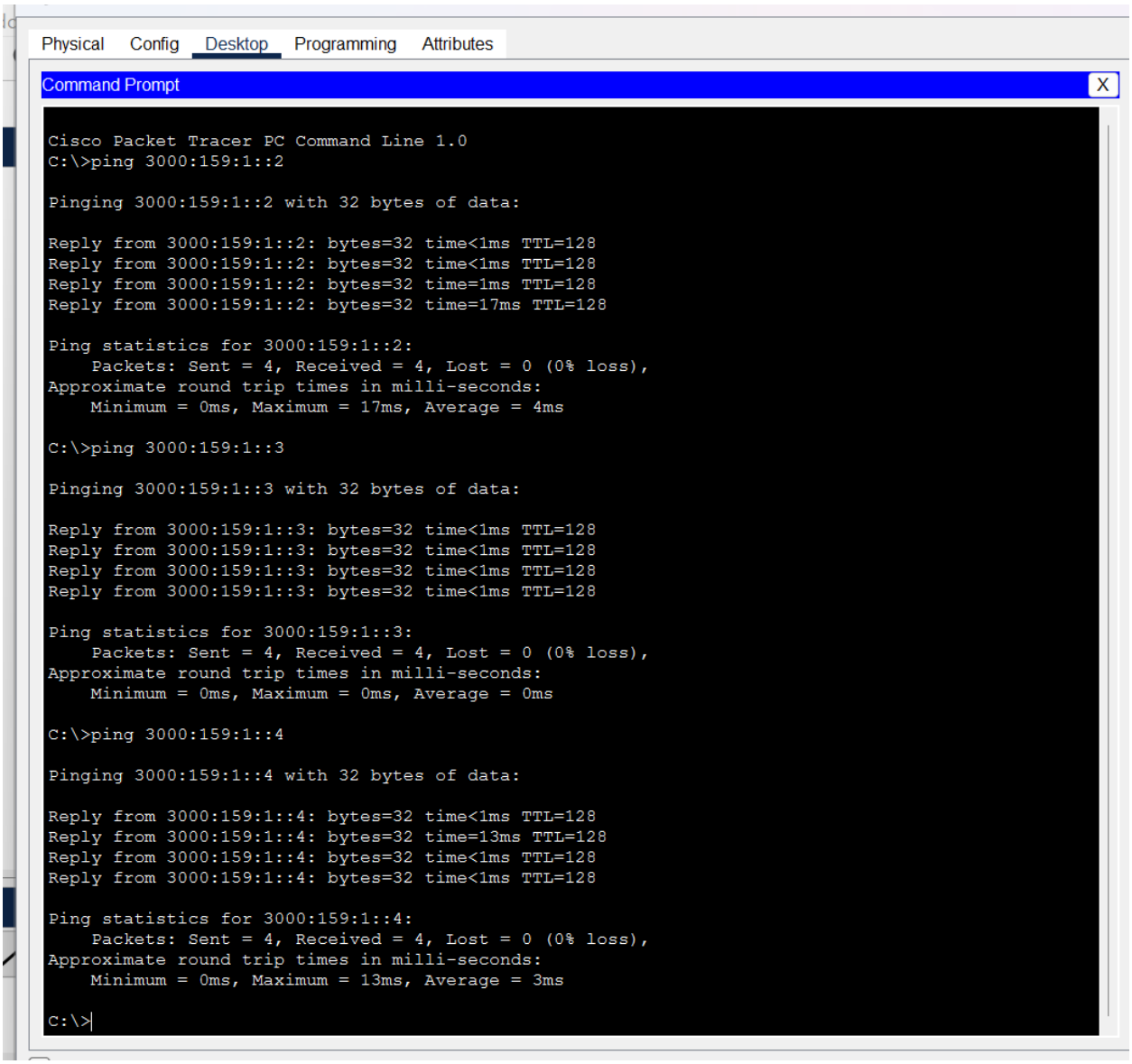



Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 3000:159:1::2
Pinging 3000:159:1::2 with 32 bytes of data:
Reply from 3000:159:1::2: bytes=32 time=1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Ping statistics for 3000:159:1::2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 3000:159:1::3
Pinging 3000:159:1::3 with 32 bytes of data:
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time=1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time=1ms TTL=128
Ping statistics for 3000:159:1::3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 3000:159:1::4
Pinging 3000:159:1::4 with 32 bytes of data:
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::4: bytes=32 time<1ms TTL=128
Ping statistics for 3000:159:1::4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 3000:159:1::5

EXERCICE 6 – Choix d'une topologie – Configuration IPv6

Topologie Etoile





CONCLUSION

Ce TD m'a permis de mettre en pratique les notions d'adressage IPv4 et IPv6 dans Cisco Packet Tracer. J'ai appris à reproduire différentes topologies, à connecter les équipements, à configurer leurs adresses IP et à contrôler la communication entre les machines avec la commande ping. Les exercices sont considérés comme réussis lorsque les adresses sont correctement configurées et que les tests de connectivité aboutissent. En cas d'échec, il faut vérifier les câbles, les ports utilisés, les adresses IP, les masques ou préfixes et la configuration de chaque équipement.